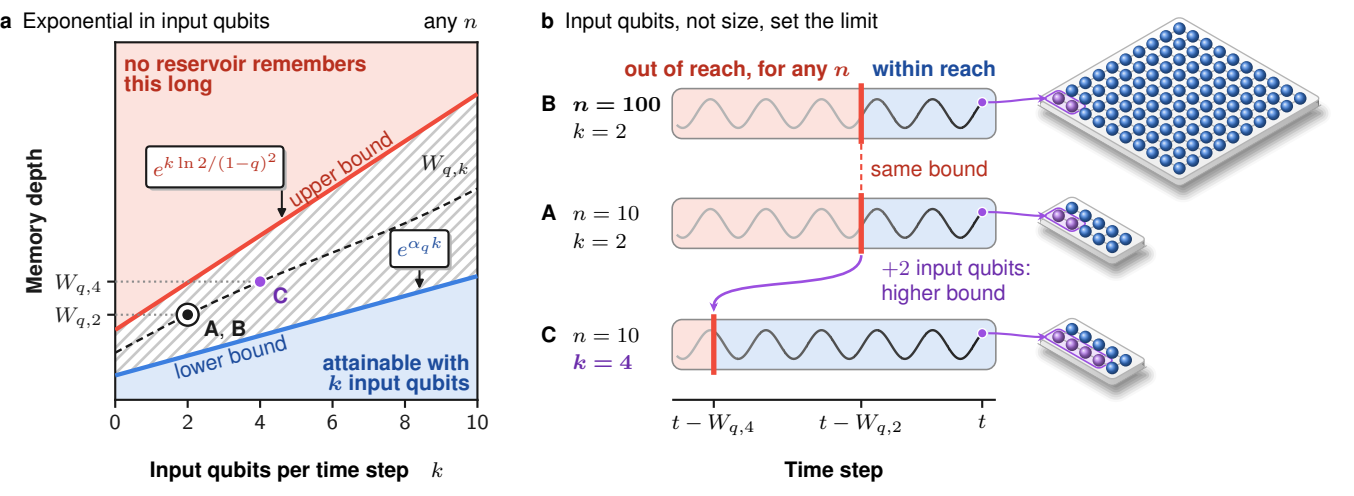

a Exponential in input qubits
any n
no reservoir remembers this long
$e^{k \ln 2/(1-q)^2}$
upper bound
$W_{q,k}$
$e^{\alpha_q k}$
C
$W_{q,4}$
$W_{q,2}$
A, B
lower bound
attainable with k input qubits
Memory depth
0
2
4
6
8
10
Input qubits per time step k
b Input qubits, not size, set the limit
out of reach, for any n
within reach
B $n = 100$
$k = 2$
same bound
A $n = 10$
$k = 2$
+2 input qubits: higher bound
C $n = 10$
$k = 4$
$t - W_{q,4}$
$t - W_{q,2}$
t
Time step